Data and Donuts Workshop

Principal component analysis for targeted PFAS concentration data

📍 URI Carother's library, Admin Suite (211) - room 203A on the second floor
📅 Wednesday, 18th of March, 2026, 01:00pm - 4:00p.m

🔗 Zoom meeting for Workshop
https://uri-edu.zoom.us/j/92305534276?pwd=wYTjo7Ybo1i0tOtpvdn3o6YrEjdV3e.1

01:00pm – 01:30pm Sample collection sheet, Code, and Miscellaneous

- Sample collection sheet
- Definitions of quality control sample types
- File naming conventions
- PFAS SCIEX Data Processing Code
- Other issues and questions

01:30pm - 2:20pm Principal Component Analysis

- Introduction to Principal Component Analysis (PCA)
- Introduction to Coding in Jupyter(Python) and R
- Principal Component Analysis for targeted PFAS concentration data

2:30pm – 4:00pm Hands-on Coding Tutorial

- As example data, we will use environmental PFAS concentrations detected in rivers, groundwater, and estuaries by former STEEP trainee Matt Dunn (https://pubs.acs.org/doi/full/10.1021/acsestwater.4c01164). If you plan to apply PCA in your own paper or study, feel free to bring your own dataset as well.
- You will only need your laptop. If you already have **R or Python** installed, you will be able to use it. Otherwise, you can work with browser-based solutions such as **webr.sh** or **Google Colab**.
- The time plan is quite ambitious. I do not expect beginners to complete the entire exercise in 1.5 hours. If you are new to programming, I recommend focusing on loading and exploring the data and getting comfortable with the programming environment rather than fully implementing PCA. I will provide example solutions after the workshop and will be available for follow-up questions.
- Drinks and donuts will be provided 😊